

# Projects

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**One project carries the summative weight of this course**, and it replaces the former midterms and final. It runs in two parts on one clinical dataset. Part 1, *One Problem, Three Lenses* (milestones 1.1 to 1.4), is due **Week 5, the project midpoint**. Part 2, *From Inference to Discovery* (milestones 2.1 to 2.5), closes it out, due **Week 7.5**.

This one project is the whole Projects component. See [Assessment](#) for how the components combine.

## One cohort, all term

**One clinical dataset carries the whole semester.** A single continuous prediction problem is examined through progressively richer lenses, then extended from prediction into discovery.

The data is a curated **NHANES 2021-2022** extract: 5,102 adults, six leakage-safe non-blood-pressure features (age, bmi, waist, chol, hdl, hba1c). Part 1 predicts systolic blood pressure from age. Part 2 binarizes blood pressure into a hypertension label under the ACC/AHA rule ( $SBP \geq 130$  or  $DBP \geq 80$ , prevalence 38.8%) and predicts it from those same six features, with the blood-pressure columns held out so the label cannot leak into its own predictors.

## Part 1: One Problem, Three Lenses

*Assigned Week 1, due Week 5 (the project midpoint). Modules 1 to 4.*

Same dataset, same single predictor (age  $\rightarrow$  risk), three accreting lenses, then a synthesis.

| Milestone | Lens                                                     | Due    | Gated by                                   |
|-----------|----------------------------------------------------------|--------|--------------------------------------------|
| 1.1       | Least squares (design matrix, cross-validation, L2)      | Week 3 | <a href="#">Checkpoint quiz 1.1</a> (Wk 2) |
| 1.2       | Maximum likelihood (Gaussian noise, predictive variance) | Week 4 | <a href="#">Checkpoint quiz 1.2</a> (Wk 3) |
| 1.3       | Bayesian (conjugate posterior, credible intervals)       | Week 5 | <a href="#">Checkpoint quiz 1.3</a> (Wk 4) |
| 1.4       | Synthesis: all three lenses side by side                 | Week 5 | structured peer review                     |

Part 1 stays single-predictor polynomial the whole way, so model order is the only complexity knob. It closes on a **closed-form conjugate Gaussian posterior**, which is exactly what Part 2 takes away.

## Part 2: From Inference to Discovery

*Begins at the midpoint (Week 5), due Week 7.5. Modules 5 to 7.*

Binarizing the outcome breaks conjugacy. The posterior goes intractable, approximate inference becomes necessary, and the feature space expands from one predictor to all six. Organized as two thematic gates plus a closing synthesis.

| Milestone | Lens                                                        | Due      | Gated by                           |
|-----------|-------------------------------------------------------------|----------|------------------------------------|
| 2.1       | Approximate inference (Newton-Raphson, Laplace, Metropolis) | Week 6   | Checkpoint quiz 2.1 (Wk 5), Gate A |
| 2.2       | Classification (soft-margin SVM from scratch)               | Week 6   | Gate A, with 2.1                   |
| 2.3       | Clustering (k-means, WCSS, restarts, choosing K)            | Week 7   | Checkpoint quiz 2.3 (Wk 6), Gate B |
| 2.4       | Dimensionality reduction (PCA, biplot)                      | Week 7   | Gate B, with 2.3                   |
| 2.5       | Synthesis: one cohort, a whole semester                     | Week 7.5 | structured peer review             |

**Gate A** (supervised) is satisfied when 2.1 and 2.2 are both Satisfactory and Checkpoint quiz 2.1 is passed. **Gate B** (unsupervised) when 2.3 and 2.4 are both Satisfactory and Checkpoint quiz 2.3 is passed. The **synthesis** milestone 2.5 is gated by peer review, not a checkpoint.

## Checkpoints

Each checkpoint is a short open-book **D2L quiz** on the module's derivations. It is named for the milestone it gates, and it is always due **the week before** that milestone, so the derivation lands before the milestone that uses it.

| Checkpoint quiz | Due    | Gates                        | Topic                             |
|-----------------|--------|------------------------------|-----------------------------------|
| 1.1             | Week 2 | Milestone 1.1 (Wk 3)         | Normal equations                  |
| 1.2             | Week 3 | Milestone 1.2 (Wk 4)         | MLE for the weights               |
| 1.3             | Week 4 | Milestone 1.3 (Wk 5)         | Posterior of a conjugate pair     |
| 2.1             | Week 5 | Milestone 2.1 (Wk 6), Gate A | Newton-Raphson update for the MAP |
| 2.3             | Week 6 | Milestone 2.3 (Wk 7), Gate B | k-means objective and PCA         |

Each quiz is 20 to 30 minutes: mostly auto-graded items that resolve on submission, plus two or three short written responses graded Satisfactory / Not-Yet with one revision. All five quizzes are required, and each gates the milestone it is named for. The numerical verification that used to sit beside these derivations is not gone, it moved where the code already lives: Homework Units 1, 3, and 4 pick up the checkpoint 1.1, 1.2, and 1.3 verification exercises, and the 2.1 and 2.3 verifications were already required inside milestones 2.1, 2.3, and 2.4. Nothing is assessed twice.

## Ground rules

- Implement every estimator **from scratch** in NumPy and SciPy. No scikit-learn or other off-the-shelf solvers.
- The info521 library is **plumbing only**: data, plotting, self-checks. It contains no fitting routines by design.
- Every milestone lists its Satisfactory criteria at the top of its own notebook. All criteria must be met. It is all-or-nothing, never partial credit.
- A **Not-Yet** returns with targeted feedback and one revise cycle.

## Getting your repo

Project materials are distributed as a repository, not as files to download. The join link is posted on **D2L** at term start. Your work lives in your own private repo; you clone it, work locally, and push.

### i Setup

Python 3.11 with numpy 2.1.x. Python 3.14 is not supported (a C-ABI incompatibility with numpy). From your repo root:

```
uv venv --python 3.11 .venv
source .venv/bin/activate
uv pip install -e .
```

Quarto must be installed separately (<https://quarto.org>); it is not a Python package.